Healthcare Readmission Analysis

Temporary Title:

Predicting Hospital Readmissions Using Patient Demographics and Medical History

Research Question:

To what extent can patient demographics and medical history predict the likelihood of hospital readmission within 30 days, and which predictive model achieves the best performance?

Introduction:

Hospital readmissions within 30 days are a major concern in the U.S. healthcare system, both in terms of cost and patient well-being. According to the Centers for Medicare & Medicaid Services (CMS), hospitals face financial penalties for excessive readmissions, making it critical to identify patients at higher risk. Predictive analytics offers the opportunity to flag vulnerable patients early, enabling healthcare providers to design interventions that reduce unnecessary readmissions.

This study aims to investigate whether patient demographic information (age, gender, race), diagnosis codes, length of stay, and previous treatment history can be used to predict hospital readmission within 30 days. By applying data mining and predictive modeling techniques, the goal is to identify the most important features that influence readmission risk.

Expected Contribution:

This research will contribute by:

- Identifying the **key factors** that drive hospital readmissions.
- Providing insights that healthcare institutions can use to reduce costs and improve patient care.

Dataset:

The project will use the UCI Diabetes 130-US hospitals dataset, which contains over 100,000 hospital admissions records across 10 years, with details on patient demographics, diagnoses, and whether they were readmitted.

Dataset Overview:

A. - Target Variable : readmitted -> Values : NO, >30, <30
B. - Variable : 50
C. - Observation : 101766
D. - Duplicate : 0
E. - Numeric Variables : 13
F. - Categorical Variables : 30
G. - Text Variables : 4
H. - Boolean Variables : 3

Variable Name	Description
encounter_id	Unique identifier of an encounter
patient_nbr	Unique identifier of a patient
race	Values: Caucasian, Asian, African American, Hispanic, and other
gender	Values: male, female, and unknown/invalid
age	Grouped in 10-year intervals: [0, 10), [10, 20),..., [90, 100)
weight	Weight in pounds.
admission_type_id	Integer identifier corresponding to 9 distinct values, for example, emergency, urgent, elective, newborn, and not available
discharge_disposition_id	Integer identifier corresponding to 29 distinct values, for example, discharged to home, expired, and not available
admission_source_id	Integer identifier corresponding to 21 distinct values, for example, physician referral, emergency room, and transfer from a hospital
time_in_hospital	Integer number of days between admission and discharge
payer_code	Integer identifier corresponding to 23 distinct values, for example, Blue Cross/Blue Shield, Medicare, and self-pay

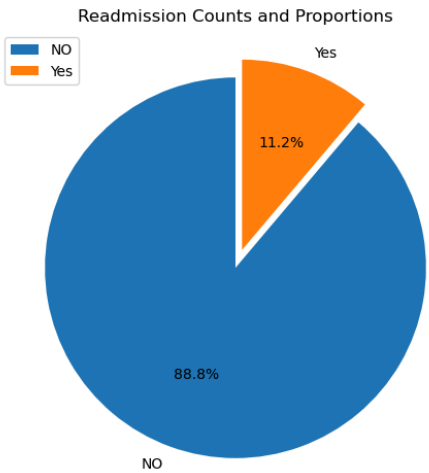
medical_specialty	Integer identifier of a specialty of the admitting physician, corresponding to 84 distinct values, for example, cardiology, internal medicine, family/general practice, and surgeon
num_lab_procedures	Number of lab tests performed during the encounter
num_procedures	Number of procedures (other than lab tests) performed during the encounter
num_medications	Number of distinct generic names administered during the encounter
number_outpatient	Number of outpatient visits of the patient in the year preceding the encounter
number_emergency	Number of emergency visits of the patient in the year preceding the encounter
number_inpatient	Number of inpatient visits of the patient in the year preceding the encounter
diag_1	The primary diagnosis (coded as first three digits of ICD9); 848 distinct values
diag_2	Secondary diagnosis (coded as first three digits of ICD9); 923 distinct values
diag_3	Additional secondary diagnosis (coded as first three digits of ICD9); 954 distinct values
number_diagnoses	Number of diagnoses entered to the system
max_glu_serum	Indicates the range of the result or if the test was not taken. Values: >200, >300, normal, and none if not measured
A1Cresult	Indicates the range of the result or if the test was not taken. Values: >8 if the result was greater than 8%, >7 if the result was greater than 7% but less than 8%, normal if the result was less than 7%, and none if not measured.
metformin	Values: up, down, steady, no

Variable Name	Description
repaglinide	Values: up, down, steady, no
nateglinide	Values: up, down, steady, no
chlorpropamide	Values: up, down, steady, no
glimepiride	Values: up, down, steady, no
acetohexamide	Values: up, down, steady, no
glipizide	Values: up, down, steady, no
glyburide	Values: up, down, steady, no
tolbutamide	Values: up, down, steady, no
pioglitazone	Values: up, down, steady, no
rosiglitazone	Values: up, down, steady, no
acarbose	Values: up, down, steady, no
miglitol	Values: up, down, steady, no
troglitazone	Values: up, down, steady, no
tolazamide	Values: up, down, steady, no
examide	Values: up, down, steady, no
citoglipton	Values: up, down, steady, no
insulin	Values: up, down, steady, no
glyburide-metformin	Values: up, down, steady, no
glipizide-metformin	Values: up, down, steady, no
glimepiride-pioglitazone	Values: up, down, steady, no
metformin-rosiglitazone	Values: up, down, steady, no

metformin-pioglitazone	Values: up, down, steady, no
change	Indicates if there was a change in diabetic medications (either dosage or generic name). Values: change and no change
diabetesMed	Indicates if there was any diabetic medication prescribed. Values: yes and no
readmitted	Days to inpatient readmission. Values: <30 if the patient was readmitted in less than 30 days, >30 if the patient was readmitted in more than 30 days, and No for no record of readmission.

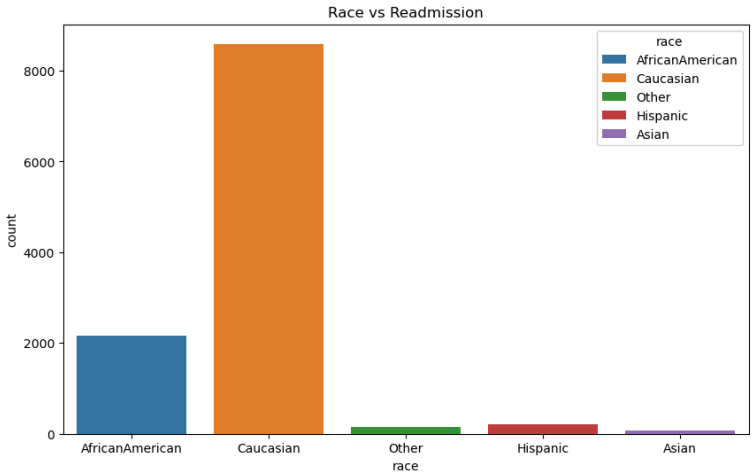
Exploratory Data Analysis

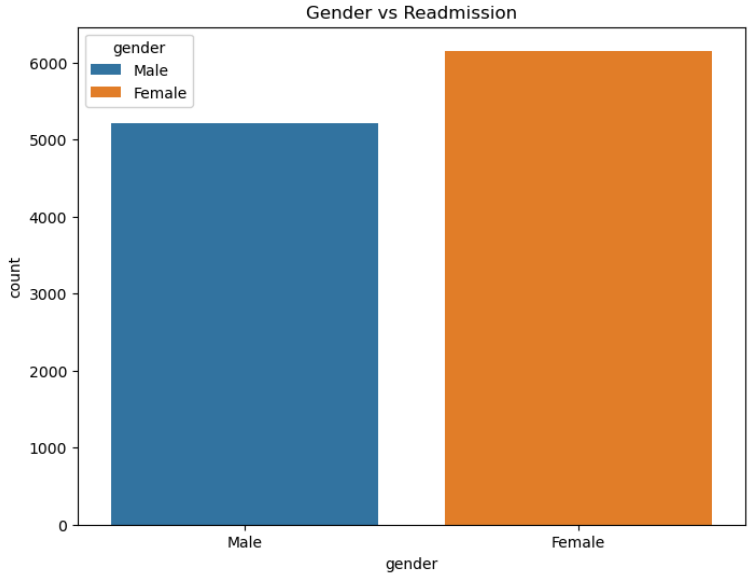
1. Readmission Count and Proportions

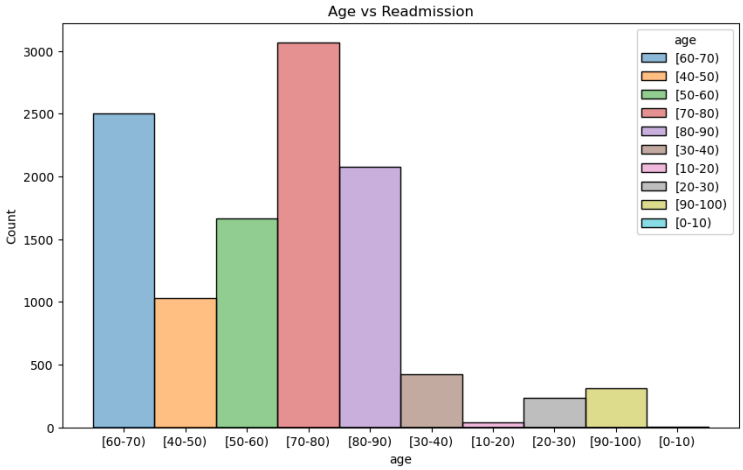


Comment: Out of the 101766 observations 11.2% are readmission cases.

2. Demographic Insights



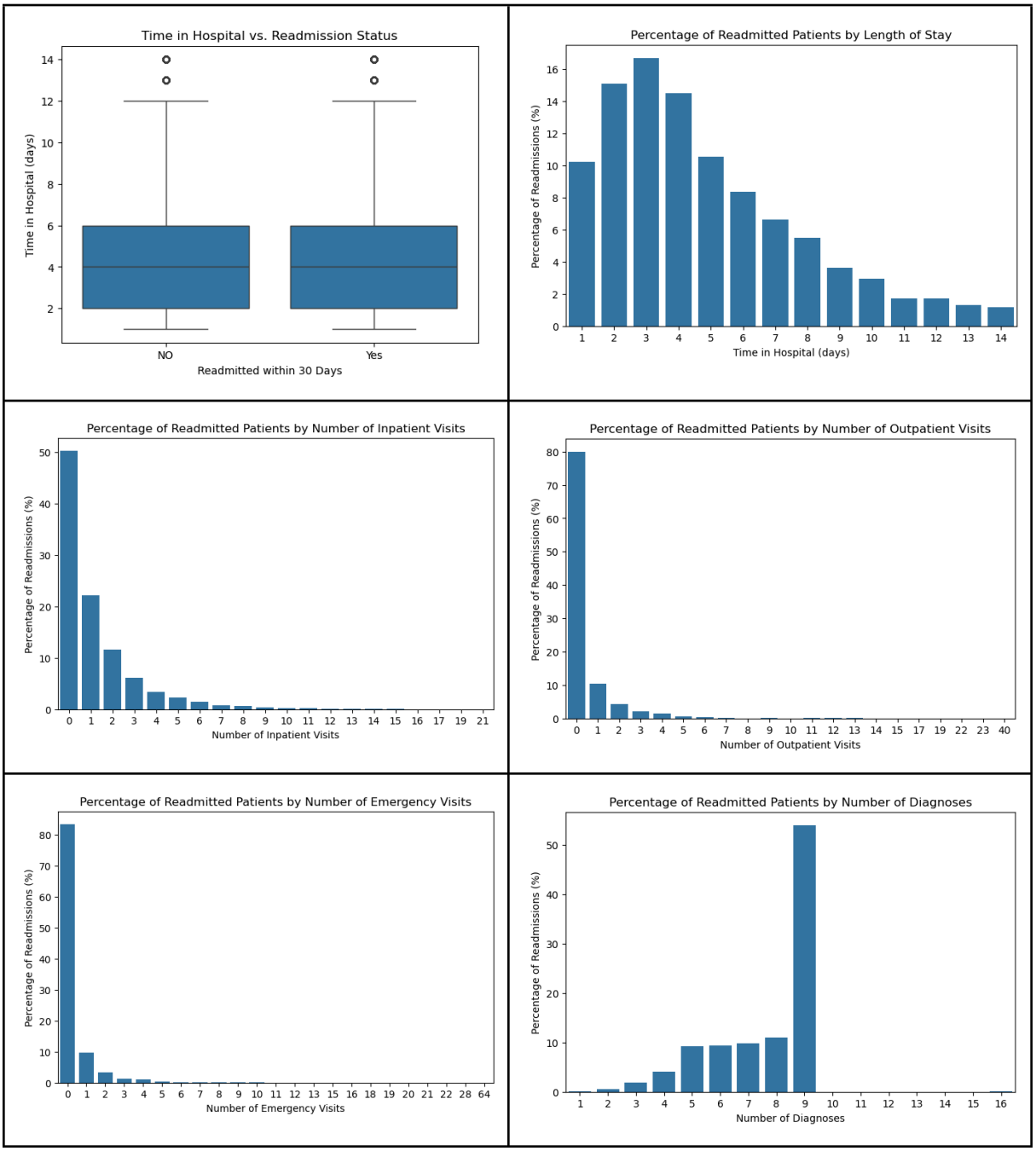




Summary of demographic Trends

Overall, the demographic analysis reveals that age and gender are significant predictors of hospital readmission risk, while race may show indirect effects depending on population composition and health disparities.

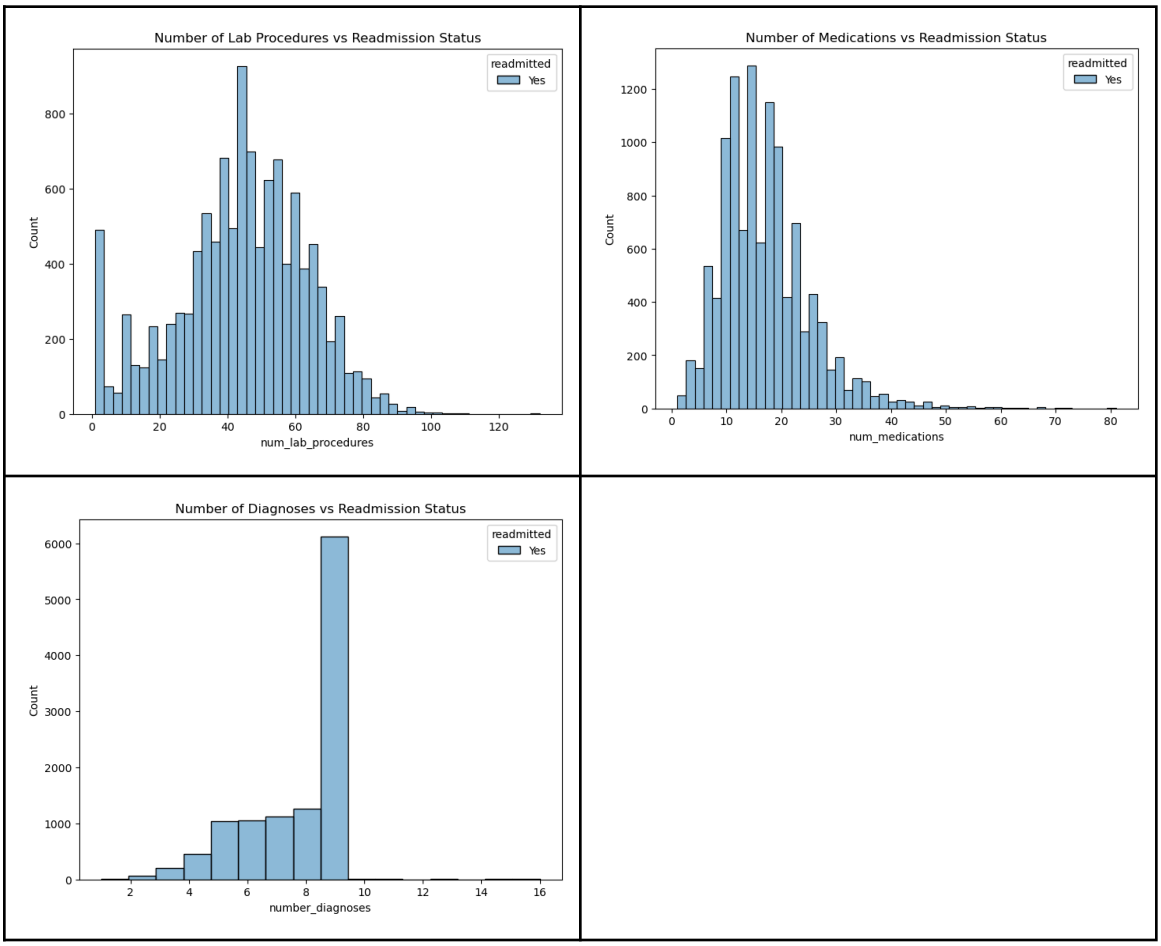
3. Hospital Stay and Visits



Summary

The comparison between readmitted and non-readmitted patients reveals that both groups have similar median hospital stays around 4 days. The difference seems small but if we focus on the readmitted patients, then the boxplot reveals that many of the readmitted patients spent less than 5 days. Maybe some patients were discharged too soon.
Other findings indicate that most readmitted patients had limited recent hospital interactions, particularly a lack of outpatient follow-up, which may contribute to preventable readmissions. The results suggest that improving continuity of care and post-discharge follow-up could reduce the likelihood of patients returning to the hospital within 30 days.

4. Lab and Meds



Summary
Overall, the analysis shows that readmitted patients tend to have greater clinical complexity, characterized by a higher number of diagnoses, medications, and laboratory tests. These findings reinforce the idea that disease burden and treatment intensity are

key factors in readmission risk. Incorporating these variables into predictive models can help identify high-risk patients who could benefit from better discharge planning and post-hospitalization follow-up.

5. Medication Management

readmitted	Yes	%
diabetesMed		
No	2246	19.776349
Yes	9111	80.223651

readmitted	Yes	%
change		
Ch	5558	48.93898
No	5799	51.06102

Summary
Readmission vs Diabetes Medication
80% of readmitted patients were on diabetes medication, while only ~20% were not. This indicates that patients undergoing diabetes treatment are significantly more likely to be readmitted.

Readmission vs Medication change
The split is nearly even (49% vs 51%), meaning readmissions occur in both groups with or without medication change.

Data Preprocessing and Feature Engineering

1. Handling Missing Values

   The dataset contained missing values primarily in weight (95%), payer_code (37%), and medical_specialty (45%).
   Variables with an excessive number of missing data and low predictive relevance such as weight and payer_code were removed. Categorical variables including race, medical_specialty, and diagnostic fields (diag_1, diag_2, diag_3) were replaced with the label “Unknown,” while laboratory result variables (max_glu_serum, A1Cresult) were filled with “None” to indicate unperformed tests. This approach ensures data integrity while retaining clinically meaningful information for modeling.

2. Encoding Binary columns and multi-category columns
   a. Binary Encoding: gender, change, diabetesMed, readmitted
   b. One-Hot Encoding: Multi-class variables (diagnoses, admission type, discharge disposition, etc.)

c. Ordinal Mapping: Medication use features (metformin, insulin, etc.) mapped to {0: 'No', 1: 'Steady', 2: 'Up', 3: 'Down'}
d. Map age ranges to numerical values

3. Addressing class imbalance
   Since only about 11% of patients were readmitted within 30 days, SMOTE (Synthetic Minority Oversampling Technique) was applied after splitting to balance the training data. Post-SMOTE, both classes contained an equal number of samples (72324 each).

Modeling

1. Data Split
   Data was divided into 80% training and 20% testing sets using stratified sampling to preserve class proportions.

2. Logistic Regression
   Logistic Regression served as the baseline due to its interpretability.

Results:

Metric	Class 0	Class 1
Precision	0.892	0.117
Recall	0.603	0.417
F1-Score	0.719	0.182
Accuracy: 0.582		

Interpretation:

The model correctly identified ~42% of readmitted patients (high recall) but with low precision. It tended to over-predict readmissions.

3. Random Forest Classifier
   A Random Forest model was trained on the balanced dataset (X_train_sm, y_train_sm).

Results:

Metric	Class 0	Class 1
Precision	0.891	0.137
Recall	0.914	0.109
F1-Score	0.902	0.121
Accuracy: 0.824		

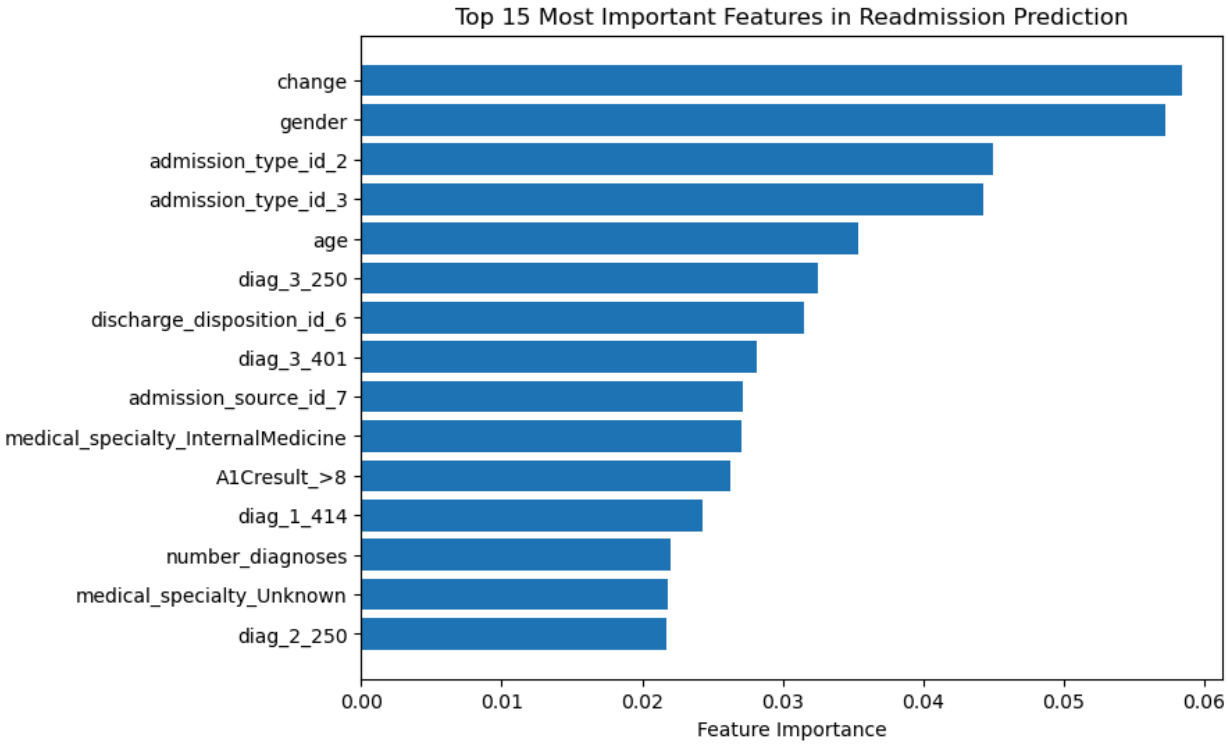
Interpretation:

The Random Forest achieved strong overall accuracy but lower sensitivity to readmissions (recall = 0.109).
This trade-off indicates it is more conservative, minimizing false positives but missing some actual readmissions.

Model Comparison

Model	Accuracy	Recall (1)	Precision (1)	F1 (1)	Comment
Logistic Regression	0.582	**0.417**	0.117	0.182	Captures more at-risk patients (higher recall).
Random Forest	**0.824**	0.109	**0.137**	0.121	Better overall accuracy, but misses some positives.

Feature Importance Analysis



An analysis of feature importance from the Random Forest model revealed that treatment-related and clinical variables played the most significant roles in predicting 30-day hospital readmissions.
The most influential feature was change, indicating whether a patient's diabetes medication regimen was altered during their hospital stay. Medication changes often signal unstable or worsening health conditions, making such patients more likely to be readmitted.

Other key predictors included A1Cresult_>8 (poor glycemic control), and admission type categories corresponding to emergency or urgent admissions — all markers of higher disease severity.
Demographic factors like age and gender also contributed meaningfully, highlighting how underlying population characteristics interact with medical complexity.

Together, these findings suggest that a combination of treatment instability, metabolic control, and chronic disease burden are the strongest predictors of hospital readmission in this dataset.

References

- Strack, B., DeShazo, J. P., Gennings, C., et al. (2014). Impact of HbA1c Measurement on Hospital Readmission Rates: Analysis of 130 US Hospitals.
- UCI Machine Learning Repository. Diabetes 130-US hospitals for years 1999–2008 Data Set.
- GeeksforGeeks. (2024). *Implementing SMOTE for Imbalanced Data in Python*. Retrieved from https://www.geeksforgeeks.org/machine-learning/smote-for-imbalanced-classification-with-python/

Additional Resources

- W3Schools. (2024). Python and Pandas Documentation. Retrieved from https://www.w3schools.com/
- GeeksforGeeks. (2024). Python Programming and Machine Learning Tutorials. Retrieved from https://www.geeksforgeeks.org/
- Codecademy. (2024). Machine Learning with Python Course. Retrieved from https://www.codecademy.com/
- UCI Dataset
- Scikit-learn docs
- Imbalanced-learn docs